

Ethan Tay

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EDUCATION

University of Washington

Bachelor of Science in Computer Science

Seattle, WA

Expected June 2028

PROJECTS

- WINGS — Calisthenics App** | *Vite, React, Firebase, Cloudflare, HTML/CSS, JavaScript* Feb 2024 – Present
- Built, deployed, and continuously iterated WINGS, an end-to-end React/Firebase progressive web application with authentication, skill tracking, and gamified progression supporting **250 daily active users (1,000 peak DAU)**
 - Grew @wings.sw to **30K followers and 3.5M+ organic views** through self-produced short-form content, driving user acquisition with no cost
 - Implemented a React/Vite frontend backed by Firebase Auth and Firestore, and integrated **React Flow** to render interactive skill trees that made progression paths easier for users to explore, increasing user engagement
- FlowState — Focus Desktop App** | *Electron, React, TypeScript, Python, AWS Cognito* May 2026 – Present
- Developed an end-to-end cross-platform **Electron desktop app** (React/TypeScript) for automated focus tracking, integrating a Python-based OCR pipeline (Tesseract) for real-time on-screen activity classification, connecting Electron and Python via IPC
 - Designed and implemented a DynamoDB backend with AWS Cognito authentication, enabling persistent user accounts, session history, and stat tracking (e.g. in-app currency, streaks)
 - Architected end-to-end user state synchronization across React frontend, Python backend, and AWS-hosted data layer, supporting real-time session tracking and productivity analytics for early users
- Real-time Hand Gesture Controller** | *Python, PyTorch, MediaPipe, PyQt6* Jan 2026 – July 2026
- Built AI hand gesture recognition pipeline: MediaPipe async hand tracking → 75-dim feature extraction → LSTM sequence classifier → OS-level cursor and keyboard control via pynput
 - Trained a PyTorch LSTM (36K parameters) on a self-labeled dataset (11,250 frames, 5 gesture classes), achieving **92% validation accuracy** with **<1ms LSTM inference latency**
 - Implemented a sliding-window temporal voting system to eliminate gesture flicker, sustaining smooth real-time performance at 30fps
- School Lost & Found Website** | *HTML/CSS, JavaScript, Firebase* Nov. 2025 – March 2026
- Built a full-stack Lost & Found platform with user authentication, admin dashboard, and image upload functionality for an business competition
 - Won **1st place regionally** (Website Coding & Development) out of all competing teams

EXPERIENCE

Robotics Programming Lead

Thunderbots Team 5604

Sep. 2024 – Dec. 2025

Vancouver, WA

- Led software for a 12-member FTC robotics team across two competition seasons, writing Java/FTC SDK teleop loops, motor-control logic, and drivetrain behavior while coordinating code updates for each season's robot design
- Maintained and extended multi-year Java codebase, adapting control logic to each season's new constraints and robot configuration

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML/CSS

Frameworks: React, Electron

Cloud & Platforms: Firebase, AWS Cognito, AWS DynamoDB, Cloudflare Pages

Developer Tools: Git, Vite, VS Code

Libraries: MediaPipe, PyTorch, Tesseract OCR